Ryan Joseph Patrick Casalino

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Summary

Senior Software & Backend Engineer with 8 years real world software engineering experience currently specializing in high-availability public infrastructure and containerized microservices. Proven track record of driving zero-downtime blue/green deployments, managing AWS infrastructure at scale, and architecting resilient systems and APIs. Private-sector experience ranges from startups focused on monetizing Twitch streams to mid-size companies focused on end-to-end payments and "big data".

Experience

Ad Hoc, LLC

Senior Software Engineer — Medicare Plan Compare API *Remote* *Feb 2022 – Present*

Backend / Platform (Go, gRPC, AWS, Terraform)

- Shipped **295 merged pull requests** across 4.5 years of sustained delivery, authoring **~49K lines** of production **Go** powering CMS's public-facing Medicare.gov Plan Compare experience before wide adoaption of GenAI tools such as Claude Code and Codex. I learned the system by working on it directly and now use these tools to aid my team and I in maintaining the system so as to avoid a great amount of typing.
- Owned end-to-end backend service architecture across **microservices**, building robust gRPC/protobuf API layers backed by PostgreSQL and orchestrated on AWS infrastructure.
- Provisioned and managed cloud infrastructure using **Terraform and Terragrunt**.
- Managed CI/CD jobs using **Groovy and Jenkins**.
- Using **Python and Bash**, streamlined internal maintenance and ETL operations. I oversaw the development to replace a CLI written in Bash to one written in Python with Click (a Python CLI library) thus reducing maintenance burden.
- Advocated for and integrated **Nix** dev-shells and build environments to ensure reproducible local development setups and eliminate "works on my machine" inconsistencies across teams.

Sinch *Bothell, WA* *2020 – 2022*

Software Engineer (Ruby, AWS, Automation)

- Engineered an automated, headless video rendering pipeline using Ruby and Bash to dynamically orchestrate AWS EC2 Windows instances executing Adobe After Effects CLI tasks.
- Maintained and refactored complex backend architectures built on Ruby frameworks (including Trailblazer), decoupling monolith dependencies to increase reliability and developer velocity.
- Scaled personalized media generation workflows to render custom video/graphic assets on-demand for high-volume SMS and MMS customer engagement campaigns.
- Documented legacy Content Management Systems and championed developer tooling standards to streamline onboarding and reduce operational friction.

Selected Projects & Independent Engineering *Seattle, WA* *2017 – 2020*

Systems & Software Engineer

- **Declarative Infrastructure Provisioning:** Authored OpenTofu and Terraform configurations alongside layered Docker images to deploy web applications and ACM SSL certificates across AWS ECS infrastructure.

- **Embedded Systems & NixOS Enthusiast:** Built custom Nix flakes and cross-compiled NixOS images for single-board computers (Broadcom/Allwinner SoCs), configuring Wayland compositors (Sway) for hardware-accelerated kiosk environments.

Technical Skills

Languages: Go, Python, Ruby, TypeScript, SQL, Bash / Shell, JavaScript

Infrastructure & DevOps: AWS (ECS, S3, Aurora), Google Cloud, Terraform, Terragrunt, Docker

Architecture & Services: gRPC, PostgreSQL, REST APIs

Education

City University of Seattle *Expected 2027*

Master of Science in Computer Science

School of the Art Institute of Chicago

Bachelor of Fine Arts